Galaxy PROJECT WORKFLOW FOUNDRY

Build better skills. Build better workflows. Keep the receipts.

A skill is compiled from notes people can read; a workflow is type-checked before any job runs; both print a receipt you can keep.

John Chilton[1], Marius van den Beek[1], Dannon Baker[2], Danielle Callan[3], Scott Cain[1], Anton Nekrutenko[1]
[1]The Pennsylvania State University, University Park, PA, USA [2]Johns Hopkins University, Baltimore, MD, USA [3]Temple University, Philadelphia, PA, USA

1 What it does

The Galaxy Workflow Foundry helps coding agents turn a paper or a Nextflow pipeline into a native, tested Galaxy workflow.

Getting a workflow right is a knowledge problem, not a syntax problem: Galaxy's tools, collections and idioms change faster than model training data. So the Foundry keeps that knowledge in a wiki people can read and review [3], compiles it into agent skills, and lets programs, not the model, decide when a workflow is right. What it can't resolve, it writes down.

Native: Galaxy's own Tool Shed tools, pinned to exact revisions, on Galaxy collections.

Skills are binaries.

KNOWLEDGE BASE
notes, schemas, command manuals and patterns, written for people to read and review

MOLD · ONE ACTION
the exact references one action needs, plus its procedure

↓ *cast: no model, same bytes*

SKILL
frozen, for the agent

PROVENANCE
a hash for every source used

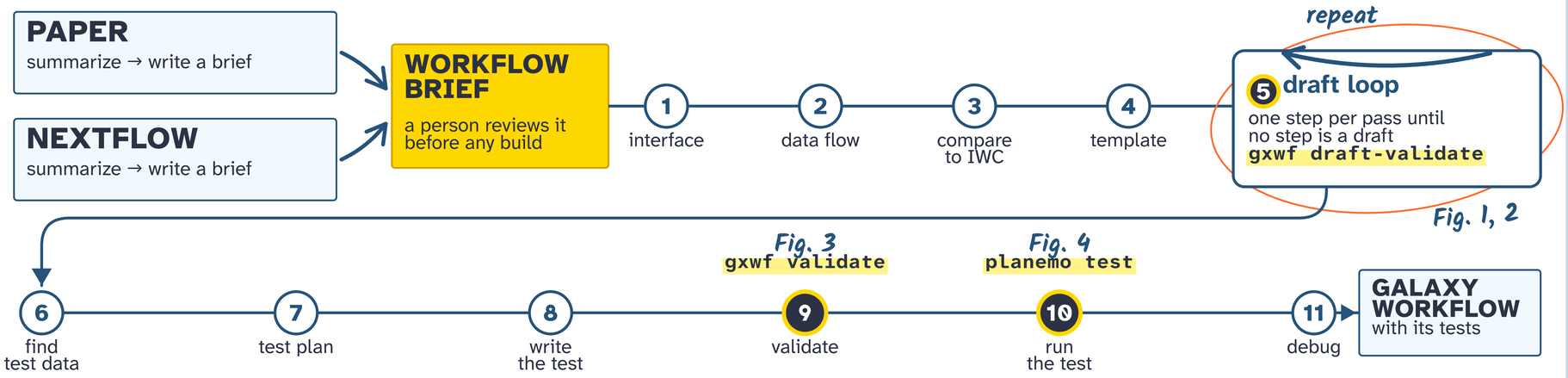
↓ *the agent runs it*

GALAXY WORKFLOW
checked by gxwf, run by Planemo

Wrong skill? Fix the note, cast again.

2 How it works

A route is a chain of Molds, each cast into a skill. A planning lane turns a paper or a Nextflow pipeline into a workflow brief. Once a person has reviewed it, one shared spine of 11 phases builds, checks and tests the Galaxy workflow. Filled stations are gates: a program, not the model, signs each one.



3 Inside the loop

Pinned, not named.

Phase 5 builds the workflow one step at a time. Each pass, `gxwf draft-next-step` picks the next unfinished step; the discover-shed-tool skill (Fig. 1) searches the Tool Shed and pins one wrapper to an exact changeset the agent fills in the step; `gxwf draft-validate` checks it against that tool's schema. The DengueSeq run took 7 passes.

4 Checked, then run

Typed, not guessed.

In phase 9, `gxwf validate` checks every parameter and every connection, collection map-over included, against the tools' real schemas. No job is queued, so an agent can repair in a tight loop.

CAST №1 09-13
`skill build · foundry @ eb2de3f1`

SKILL.md	discover-shed-tool
MOLD rev 5	05c99cba
5 refs · 1 upfront · 4 on demand	

SIGNED **`foundry-build cast`**

Figure 1 · CAST. The Tool Shed search skill the DengueSeq run used in its draft loop: Mold revision 5, 5 references, no model in the build.

PINNED №2 10-20 22:51–23:39
`Tool Shed changesets · phase 5`

devteam/**bwa_mem**	0.7.19+galaxy1 @c598cbb8b019
iuc/**samtools_view**	1.22+galaxy2 @5cf9fcc93508
iuc/**ivar_trim**	1.4.4+galaxy1 @6606a8c97889
iuc/**ivar_consensus**	1.4.4+galaxy0 @1c5448133894
bgruening/**tp_sed_tool**	9.11+galaxy0 @30b60202d990
iuc/**ivar_variants**	1.4.4+galaxy0 @5e5dfc9e71e2
iuc/**bedtools_genomecoveragebed**	2.31.1+galaxy0 @d57471728f7b

SIGNED **`gxwf tool-revisions`**

Figure 2 · PINNED. The seven tools the run pinned, as `gxwf tool-revisions` reported them. Planemo later installed exactly these.

VALIDATED №3 10-21 00:00
`replayed · gxwf 1.15.0 · 2026-10-30`

```
$ gxwf validate
  galaxy-workflow.gxwf.yml
  --connections --strict --json
Tool state . . 7 ok · 0 fail · 0 skip
Connections . . . . . 12 ok · 0 invalid
map-over . . . . all 7 steps over list
wall time . . . . . < 1 s
```

SIGNED **`gxwf validate`**

Figure 3 · VALIDATED. Phase 9's check, replayed.

The last check is a real run.

gxwf checks the structure. Then Planemo installs the pinned tools in a real Galaxy and runs the workflow's test. That run is the validation: real jobs, real outputs, checked against the expected ones.

TESTED №4 10-21 00:33
`planemo 0.75.47 · its JSON report`

```
planemo test . . . . . success
tests . . . 1 · failures 0 · errors 0
Tool Shed repos, at the pins . . . . 7
containers . . . . . 6
wall time . . . . . 11 min
```

PASS

JOBS RUN . . . . . 7 ok · 0 failed

SIGNED **`planemo test`**

Figure 4 · TESTED. Planemo's report of that run. One larger test case (about 178 MB of reads) was not run.

5 Rails, not reminders.

The companion Orbit study ran agents on public Galaxy [1]. Its lesson: skills are guidance, not enforcement [1]. What held were the rails. A connection check caught an agent's bad `tximport` wiring before anything ran, and Orbit's evidence gate now refuses by default [1].

The Foundry is built the same way: the model drafts and repairs; gxwf and Planemo decide. Its compilation pipeline is described in [1]; here we show its knowledge base, provenance and checks. [1] also reports a 60-step workflow the Foundry compiled from an RNA-seq paper's methods text; comparing with the authors' data found a model choice the methods left out.

What's inside

420 notes · **56** Molds · **54** patterns · **69** skills

One note can feed many skills: fix it once, recast, and they all get the fix.

Gaps are written down.

The run's ledger closed 14 of 19 requirements and left 5 open, each stated in words for the reviewer: for example, that the source's filtered variant table is not reproduced, a cut the reviewed brief chose.

6 Get started

The Foundry

Preprint [1]

Claude Code
```
/plugin marketplace add galaxyproject/foundry
/plugin install foundry-skills@galaxy-workflow-foundry
```

Codex
```
codex plugin marketplace add galaxyproject/foundry
codex plugin add foundry-skills@galaxy-workflow-foundry
```

gxwf, the checker
```
npm install -g @galaxy-tool-util/cli
```

From a paper: brief, review it, then build
```
/foundry-skills:pipeline-paper-to-workflow-brief
/foundry-skills:pipeline-workflow-brief-to-galaxy
```

Knowledge-base counts: galaxyproject/foundry main, 2026-10-24. Figures 1–4 and the ledger: Foundry run dengueseq-denv3-galaxy (2026-10-20), a Galaxy translation of DengueSeq [2]. Times UTC.

[1] Baker D, ..., Nekrutenko A. Verifiable Agentic Science on Public Infrastructure. bioRxiv (2026). doi:10.64898/2026.10.26.761864
[2] Vogels CBF, Hill V, et al. DengueSeq. BMC Genomics 25:433 (2024).
[3] Karpathy A. LLM wiki. gist.github.com/karpathy (2026).